1 Deep Reinforcement Learning Algorithms

Algorithm	Family	Main Idea	Core Equation	Policy	Model
Greedy	Bandits	Always choose best action	$A_t = \arg\max_a Q_t(a)$	On	Free
ϵ-Greedy	Bandits	Random exploration	$A_t = \begin{cases} \arg\max_a Q(a), & 1-\epsilon \\ \text{random}, & \epsilon \end{cases}$	On	Free
UCB	Bandits	Optimism under uncertainty	$A_t = \arg\max_a \left(Q(a) + c\sqrt{\frac{\log t}{N(a)}}\right)$	On	Free
Thompson Sampling	Bandits	Sample from posterior	$a = \arg\max_a \tilde{q}(a), \quad \tilde{q}(a) \sim P(q(a))$	On	Free
Gradient Bandits	Policy-based	Learn action preferences	$\pi(a) = \frac{e^{H(a)}}{\sum_b e^{H(b)}}$	On	Free
Dynamic Programming	Value-based	Solve known MDP	Bellman equations	Either	Based
Policy Iteration	DP	Evaluate then improve policy	$\pi'(s) = \arg\max_a q_\pi(s,a)$	Either	Based
Value Iteration	DP	Direct Bellman optimization	$V(s) \leftarrow \max_a [r + \gamma V(s')]$	Either	Based
Monte Carlo	Value-based	Learn from episodes	$V(s) \leftarrow V(s) + \alpha(G_t - V(s))$	On	Free
Temporal Difference	Value-based	Bootstrap estimates	$V(s) \leftarrow V(s) + \alpha[r + \gamma V(s') - V(s)]$	On	Free
SARSA	Value-based	Use executed actions	$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)]$	On	Free
Q-Learning	Value-based	Learn optimal values	$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$	Off	Free
DQN	Deep RL	Neural network Q-function	$L = (y - Q(s,a;\theta))^2$	Off	Free
REINFORCE	Policy Gradient	Direct policy optimization	$\theta \leftarrow \theta + \alpha G_t \nabla_\theta \log \pi_\theta(a\|s)$	On	Free
Actor-Critic	Hybrid	Actor acts, critic evaluates	$\delta = r + \gamma V(s') - V(s)$	On	Free
A2C/A3C	Actor-Critic	Use advantage estimates	$A(s,a) = Q(s,a) - V(s)$	On	Free
TRPO	Policy Gradient	Constrained updates	$\max E[r_t A_t] \quad s.t. \quad D_{KL} \leq \delta$	On	Free
PPO	Policy Gradient	Clipped optimization	$L = \min(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)A_t)$	On	Free
DDPG	Actor-Critic	Continuous actions	$\nabla_\theta J = \nabla_a Q(s,a) \nabla_\theta \mu_\theta(s)$	Off	Free
TD3	Actor-Critic	Twin critics	$y = r + \gamma \min(Q_1, Q_2)$	Off	Free
SAC	Actor-Critic	Reward + entropy	$J = E[R] + \alpha H(\pi)$	Off	Free
Dyna-Q	Hybrid	Model + simulation	Q-learning + planning	Off	Based
Prioritized Sweeping	Planning	Update important states	Bellman error priority	Off	Based
AlphaZero	Planning	Self-play + MCTS	$U(s,a) = Q(s,a) + cP(s,a)\frac{\sqrt{N}}{1+n(s,a)}$	Self-play	Based
MuZero	Planning	Learn world model	Learn networks (h, g, f)	Self-play	Learned
Dreamer	World Models	Imagine trajectories	Optimize imagined returns	Either	Based
MPC	Planning	Replan every step	$\max_{a_{0:H}} \sum_t r_t$	N/A	Based
Random Shooting	MPC	Random trajectories	$\arg\max_i R_i$	N/A	Based
CEM	Optimization	Fit elite samples	Gaussian update	N/A	Based
iLQR	Control	Linearize dynamics	Riccati recursion	N/A	Based

2 Fundamental Reinforcement Learning Equations

2.1 Return

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \tag{1}$$

The return represents the discounted cumulative future reward.

2.2 Bellman Expectation Equation

$$v_\pi(s) = \sum_a \pi(a|s) \left[r(s,a) + \gamma \sum_{s'} P(s'|s,a) v_\pi(s') \right] \tag{2}$$

2.3 Bellman Optimality Equation

$$v^*(s) = \max_a \left[r(s,a) + \gamma \sum_{s'} P(s'|s,a) v^*(s') \right] \tag{3}$$

$$q^*(s,a) = r + \gamma \sum_{s'} P(s'|s,a) \max_{a'} q^*(s',a') \tag{4}$$

3 RL Family Tree

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Reinforcement Learning
|
|-- Value-Based
|   |-- TD
|   |-- SARSA
|   |-- Q-Learning
|   |-- DQN
|
|-- Policy-Based
|   |-- REINFORCE
|   |-- TRPO
|   |-- PPO
|
|-- Actor-Critic
|   |-- A2C/A3C
|   |-- DDPG
|   |-- TD3
|   |-- SAC
|
|-- Model-Based
    |-- Dyna-Q
    |-- MPC
    |-- AlphaZero
    |-- MuZero
    |-- Dreamer
```